

Andi Zhang

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Education

University of Cambridge

PhD in Computer Science

Oct. 2019 - Mar. 2024

- Supervised by Dr. Damon Wischik. Study Bayesian neural network, Probabilistic generative models and applications about robustness of deep learning models (adversarial examples, OOD detection).

MAst in Mathematical Statistics (Part III)

Oct. 2018 - Jun. 2019

- Mathematical Tripos Part iii. Relevant modules:
Statistical Learning in Practice, Convex Optimization, Astrostatistics
Bayesian Modelling and Computation, Modern Statistical Methods

MPhil in Advanced Computer Science

Oct. 2017 - Jun. 2018

- Relevant modules:
Machine Learning for Natural Language Processing, Algebraic Path
Deep Learning for Natural Language Processing, Advanced Functional Programming
Probabilistic Machine Learning

University of Manchester

BSc (Hons) in Computer Science and Mathematics

Sep. 2014 - Jun. 2017

- Relevant modules:
Machine Learning, Quantum Computing, Compilers
Convex Optimization, Probability, Statistical Methods
Functional Analysis, Topology, Number Theory

Experience

Kirin AI Group, Huawei

Cambridge, United Kingdom

Research Intern (Part-time) | Mentor: Mr. Arnau Raventos

Nov. 2021 - Oct. 2022

- Worked on video frame interpolation and AI global illumination.

Noah's Ark Lab, Huawei

Shenzhen / Beijing, China

Research Intern | Mentor: Dr. Yitong Sun

Sep. 2020 - Sep. 2021

- Worked on deep learning uncertainty, out-of-distribution detection and lossless compression.

University of Cambridge

Cambridge, United Kingdom

Course Supervisor (Teaching Assistant)

Oct. 2019 - Jun. 2023

- Supervised IA Algorithms (2020 Lent, 2023 Lent), IA Machine Learning in Real World (2020 Lent), IA Introduction to Probability (2020 Easter, 2021 Easter), IB Further Java (2020 Michaelmas), IB Data Science (2022 Michaelmas), IB Further Graphics (2022 Michaelmas) and II Machine Learning and Bayesian Inference (2023 Lent)

MRC Biostatistics Unit, University of Cambridge

Cambridge, United Kingdom

Research Assistant | Supervisor: Dr. Sofia Villar and Dr. David Robertson

Jul. 2019 - Sep. 2019

- Built a simulation system for multi-armed bandit problem (MABP) in clinical trial.
- Presented a new test statistic for binary response MABP.

Tencent (WeChat AI)

Beijing, China

R & D Intern | Mentor: Dr. Jie Zhou

Jul. 2018 - Sep. 2018

- Concentrated on reinforcement learning and evolution strategy.

Amazon (Alexa)

Cambridge, United Kingdom

Applied Scientist Intern | Mentor: Dr. Emilio Monti

Jun. 2017 - Sep. 2017

- Concentrated on Neural Machine Translation (Sequence to sequence model).

Robotics Institute, Carnegie Mellon University

Pittsburgh, United States

Student Intern | Supervisor: Dr. Stelian Coros

Jun. 2016 - Sep. 2016

- Assisted in implementing general stance controller for the 3D printable robotic creatures.
- Helped reproduce the prototype animation system.

Publications

- Falsehoods that ML researchers believe about OOD detection** NeurIPS 2022 Workshop
Andi Zhang, Damon Wischik
In ML Safety Workshop, 36th Conference on Neural Information Processing Systems
- Out-of-Distribution Detection with Class Ratio Estimation** NeurIPS 2022 Workshop
Mingtian Zhang, Andi Zhang*, Tim Z. Xiao, Yitong Sun, Steven McDonagh*
In ML Safety Workshop, 36th Conference on Neural Information Processing Systems
- On the Out-of-distribution Generalization of Probabilistic Image Modelling** NeurIPS 2021
Mingtian Zhang, Andi Zhang*, Steven McDonagh*
In Neural Information Processing Systems annual meeting, 2021
- Measuring Uncertainty in Signal Fingerprinting with Gaussian Processes Going Deep** IPIN 2021
Ran Guan, Andi Zhang, Mengchao Li, Yongliang Wang
In International Conference on Indoor Positioning and Indoor Navigation, 2021
- Towards dynamic and scalable active learning with neural architecture adaption for object detection** BMVC 2021
Fuhui Tang, Dafeng Wei, Chenhan Jiang, Hang Xu, Andi Zhang, Wei Zhang, Hongtao Lu, Chunjing Xu
In The British Machine Vision Conference, 2021
- Solipsistic Reinforcement Learning** ICLR 2021 Workshop
Mingtian Zhang, Peter Noel Hayes*, Tim Z. Xiao, Andi Zhang, David Barber*
In Self-Supervision for Reinforcement Learning Workshop - International Conference on Learning Representations, 2021

Honers & Awards

- Top Reviewer (10%) of AISTATS 2023** Feb. 2023
- The Williams/Kilburn Medal for outstanding final year student** Jul. 2017
School of Computer Science, University of Manchester
The medal is awarded on an occasional basis to students of exceptional distinction whose performance throughout their undergraduate course has been of outstanding merit.
- University awards for outstanding academic achievement** Jul. 2017
University of Manchester
The University will consider the top 0.5% of undergraduates (nominated by each School) for this award.